

Why a dot product is just agreement

DOT PRODUCT = HOW MUCH DO THEY AGREE?



same way

+1



right angles

0



opposite

-1

Same direction, no direction, opposite direction

A dot product asks one question: **are these two arrows pointing the same way?**

- 1 Two vectors pointing the **same way** give a big positive number.
- 2 **At right angles** gives zero — no agreement at all.
- 3 **Opposite ways** gives a negative number. That is the whole intuition.

REMEMBER

Every similarity score in AI — search, recommendations, embeddings — is this one question.

